

Q.1 Find the largest sum of any two elements

```
function maxPairSum(arr) {  
    arr.sort((a, b) => b - a);  
    return arr[0] + arr[1];  
}  
  
console.log(maxPairSum([1, 9, 2, 8]));  
=> 17
```

Explanation:

- Sort array descending sum of first two numbers.

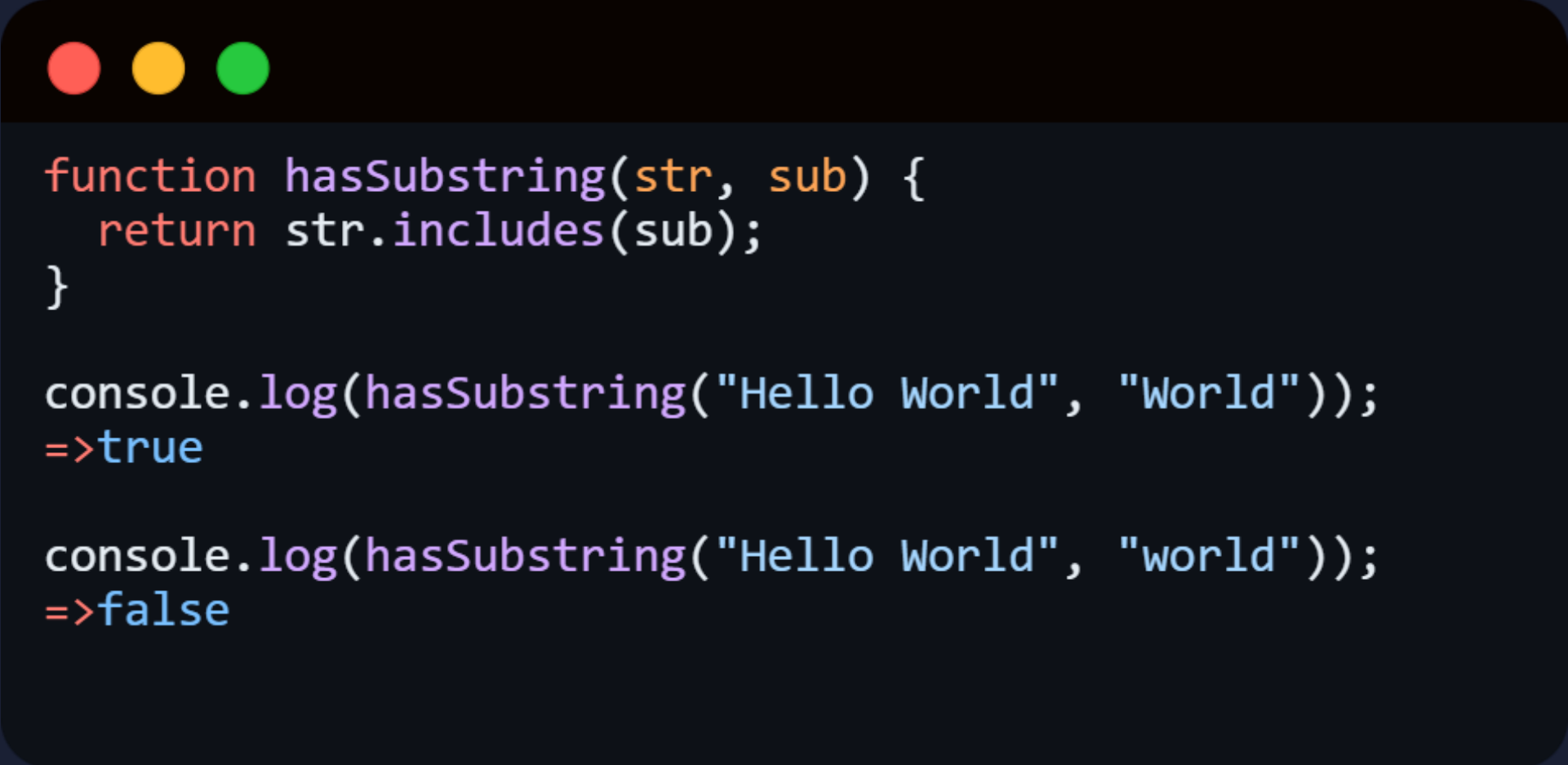
Q2. Check whether any two numbers in an array sum to a given number

```
function hasPairWithSum(arr, sum) {  
  const set = new Set();  
  
  for (let num of arr) {  
    if (set.has(sum - num)) return true;  
    set.add(num);  
  }  
  
  return false;  
}  
  
console.log(hasPairWithSum([1, 2, 4, 9], 8)); => false  
console.log(hasPairWithSum([1, 2, 4, 4], 8)); => true
```

Explanation:

- Use a Set to track numbers needed to reach the target sum.

Q3. Match substring of a string



```
function hasSubstring(str, sub) {  
  return str.includes(sub);  
}  
  
console.log(hasSubstring("Hello World", "World"));  
=>true  
  
console.log(hasSubstring("Hello World", "world"));  
=>false
```

Explanation:

- includes() checks if substring exists.
- Use toLowerCase() for case-insensitive match if needed.

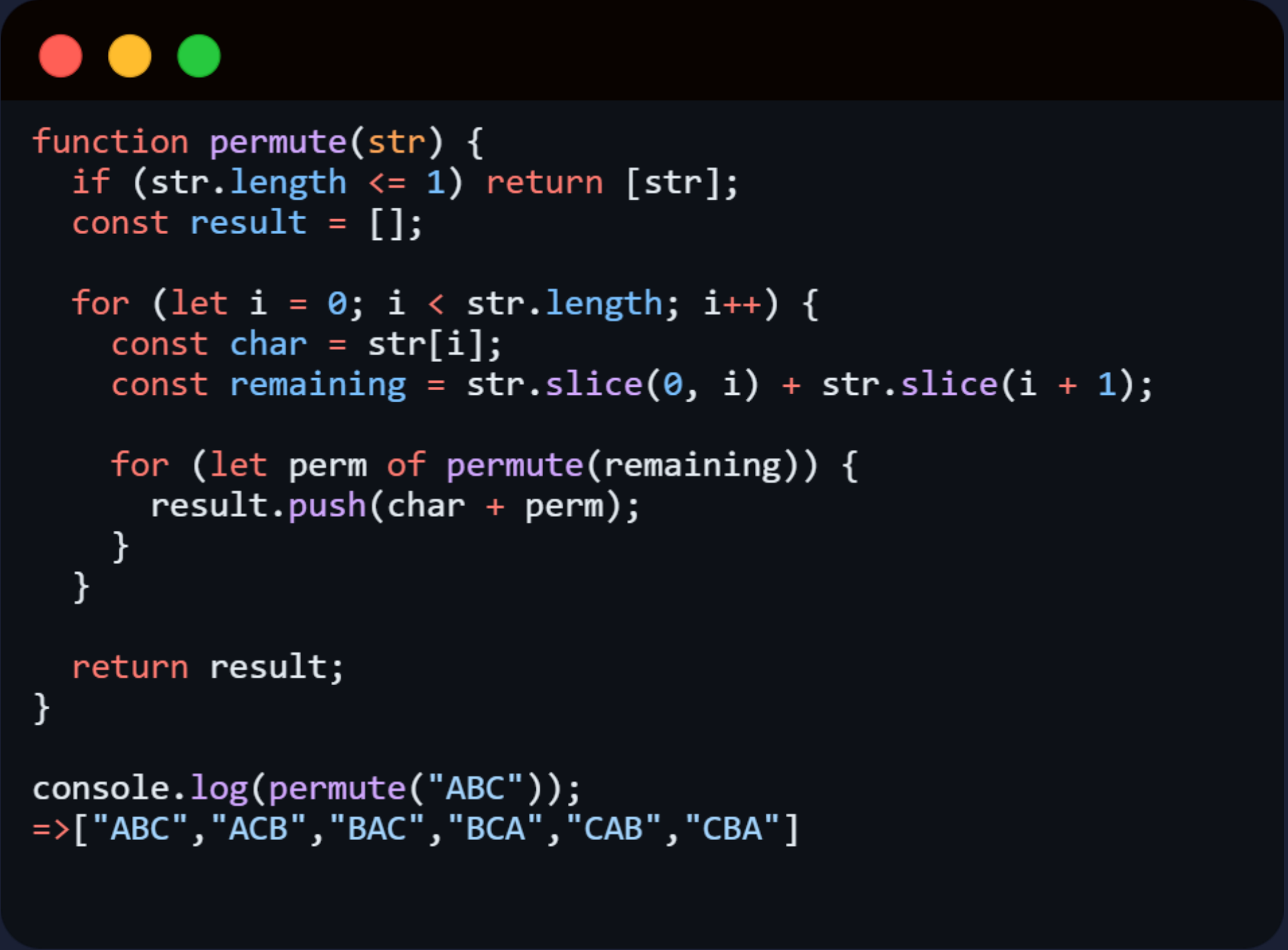
Q4. Find missing number from an unsorted array

```
function findMissing(arr, n) {  
  let total = (n * (n + 1)) / 2;  
  let sum = arr.reduce((a, b) => a + b, 0);  
  return total - sum;  
}  
  
console.log(findMissing([3, 7, 1, 2, 8, 4, 5], 8));  
=> 6
```

Explanation:

- Sum of 1 to n = $n*(n+1)/2$
- Subtract sum of array → missing number.

Q5. Create all permutations of a string



```
function permute(str) {  
  if (str.length <= 1) return [str];  
  const result = [];  
  
  for (let i = 0; i < str.length; i++) {  
    const char = str[i];  
    const remaining = str.slice(0, i) + str.slice(i + 1);  
  
    for (let perm of permute(remaining)) {  
      result.push(char + perm);  
    }  
  }  
  
  return result;  
}  
  
console.log(permute("ABC"));  
=>["ABC", "ACB", "BAC", "BCA", "CAB", "CBA"]
```

Explanation:

- Recursively fix each character and permute the rest.

Q6. Total number of zeros from 1 up to n

```
function countZeros(n) {  
  let count = 0;  
  
  for (let i = 1; i <= n; i++) {  
    count += i.toString().split('0').length - 1;  
  }  
  
  return count;  
}  
  
console.log(countZeros(20));  
=> 2
```

Explanation:

- Convert each number to string, count '0's, sum them up.